

STAT 5700

Philosophical and Ethical Issues in Statistics

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	Data Scientist (32.0 / 100)	22.1 / 100
SPECIALIZATION On-Campus Elective		

22

CURRENT AVG SCORE
across 6 job roles

64

POTENTIAL SCORE
with recommended additions

+48

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces students to philosophical issues that arise in statistical theory and practice. Topics include interpretations of probability, philosophical paradigms in statistics, inductive inference, causality, reproducible, and ethical issues arising in statistics and data analysis.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	32.0	68.7	+36.7	HIGH
Machine Learning Engineer	25.7	72.1	+46.4	HIGH
Applied Scientist	21.3	69.7	+48.4	HIGH
AI Engineer	18.7	63.6	+44.9	HIGH
ML Engineer	18.2	52.9	+34.7	HIGH
MLOps Engineer	16.8	54.2	+37.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

3 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Data Scientist

32.0 → 68.7 (+36.7 pts)

presentation social people problems

Machine Learning Engineer

25.7 → 72.1 (+46.4 pts)

learning tools exploratory design development science models model

Applied Scientist

21.3 → 69.7 (+48.4 pts)

computational research interdisciplinary problems models technical department learning

AI Engineer

18.7 → 63.6 (+44.9 pts)

robotics models learning technical model systems teams problems

ML Engineer

18.2 → 52.9 (+34.7 pts)

media learning machine systems models model world problems

MLOps Engineer

16.8 → 54.2 (+37.4 pts)

model systems learning machine media tools models development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.